

CONTEXTUAL AREAS

Bullwhip Effect Measurement and Its Implications

Li ChenFuqua School of Business, Duke University, Durham, North Carolina 27708, li.chen@duke.edu**Hau L. Lee**Graduate School of Business, Stanford University, Stanford, California 94305, haulee@stanford.edu

The bullwhip effect, or demand information distortion, has been a subject of both theoretical and empirical studies in the operations management literature. In this paper, we develop a simple set of formulas that describe the traditional bullwhip measure as a combined outcome of several important drivers, such as finite capacity, batch-ordering, and seasonality. Our modeling framework is descriptive in nature as it features certain plausible approximations that are commonly employed in practical inventory systems. The results are nonetheless compelling and can be used to explain various conflicting observations in previous empirical studies. Building on the theoretical framework, we discuss the managerial implications of the bullwhip measurement. We show that the measurement can be completely noninformative about the underlying supply chain cost performance if it is not linked to the operational details (such as decision intervals and leadtimes). Specifically, we show that an aggregated measurement over relatively long time periods can mask the operational-level bullwhip. In addition, we show that masking also exists under product or location aggregation in some illustrative cases.

Subject classifications: bullwhip effect; finite capacity; batch-ordering; seasonality; supply chain management; data aggregation.

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1. Introduction

The supply chain bullwhip effect, as described by Lee et al. (1997a), is “the phenomenon where orders to the supplier tend to have larger variance than sales to the buyer (i.e., demand distortion), and the distortion propagates upstream in an amplified form (i.e., variance amplification).” This phenomenon has been observed in many supply chains (see a detailed survey in the literature review section). However, there have also been reports of the absence of bullwhip effect in both simulated and empirical studies (Baganha and Cohen 1998, Ghali 1987, Cachon et al. 2007). In this work, we expand the existing theory to provide scientific explanation for the different bullwhip observations made in previous studies. We also discuss how different measures of the bullwhip effect are related to supply chain cost performance.

We first derive a general bullwhip formula based on a capacitated batch-order system with state-dependent stochastic demand. We believe this model setting captures the essence of most real-world scenarios. From the formula, one can obtain an integrated view of the impacts of various factors on the bullwhip measure, including demand process characteristics, system capacity, and batch-ordering. When there is no batch-ordering, our result indicates that imposing a finite capacity to the system has a smoothing effect on the order variability. This result can be further extended to show that the order receipt sequence (or shipment sequence), constrained by the upstream order-fulfillment capacity, is less

variable than the original order sequence. As such, bullwhip measurement based on the shipments (material flow) might be different from the measurement based on the orders (information flow). A detailed delineation of these two types of measurement is provided in the ensuing section. When seasonality is present in demand process, we show that including seasonality in the measurement will mask the bullwhip effect if the variability of seasonality dominates the deseasonalized demand variability. This result provides a theoretical support for the production smoothing effect observed by Ghali (1987) and Cachon et al. (2007). For a quick reference, we summarize the insight obtained in this study on the bullwhip measurement in Table 1.

Subsequently, we discuss the linkage between the bullwhip measure and the supply chain cost performance. We focus our analysis on the upstream—the receiving end of the bullwhip effect. Our results suggest that the bullwhip effect should be measured at the appropriate time unit for cost assessment purposes. For example, to assess its impact on the upstream warehouse staffing and truck fleet capacity costs, the bullwhip effect should be measured at the upstream order-fulfillment time interval. We further show that the upstream inventory cost is a function of the bullwhip measure conditional on the available information set (such as forecast updates; see, for example, Aviv 2007, Chen and Lee 2009). In this case, the bullwhip effect should be measured at the upstream inventory risk exposure period

Table 1. Summary of various influencing factors for bullwhip measurement.

Influencing factors	Amplifying	Dampening	Masking
Finite system capacity		✓	
Batch-ordering	✓		
Seasonality included in measure		✓	
Time aggregation			✓
Product/location aggregation			Inconclusive*

*Masking the bullwhip in some illustrative cases.

(i.e., the reorder period plus inventory replenishment lead time). Moreover, the conditional bullwhip measure in this case is an uncertainty-based bullwhip measure rather than the traditional variability-based bullwhip measure. The traditional measure is appropriate for assessing the inventory-related cost if the upstream party has no access or does not utilize available demand information to improve its inventory replenishment decisions. We feel this case might still apply to many small and medium-size companies, where they lack the resource in investing in advanced information technology. For a detailed discussion on the different implications between uncertainty propagation and variability propagation along the supply chain, we refer the readers to Aviv (2001, 2003, 2007) and Chen and Lee (2009).

Bullwhip Effect Measure

There are two primary definitions of bullwhip effect measurement used in the literature. Lee et al. (1997a) originally described the bullwhip effect as a form of “information distortion” and measured it by comparing the order variance with the demand variance (where order can also be interpreted as production release in a manufacturing setting). This definition captures the distortion of *information flow* that goes upstream (the downstream stage’s order is the demand input to the upstream stage). A second definition, used in most empirical studies, compares the variance of order receipts (or shipments) with the variance of sales. In some cases, the order receipt information, if not available, is inferred from the sales and inventory data (see Blinder 1981, Cachon et al. 2007). This definition essentially captures the distortion of *material flow* that goes downstream.

The bullwhip measurements based on these two definitions are usually good approximation to each other (as material flow more or less follows information flow), but they differ in concept. The information-based definition has a direct linkage to supply chain cost because the upstream inventory/capacity decision is driven by the downstream order information. Hence, the information-based bullwhip effect is a cost driver. In contrast, order receipts information is the *outcome* of the upstream order-fulfillment decision process, rather than an *input* to the decision process. Hence, the material-based bullwhip effect is the consequence of

the information-based bullwhip effect. Moreover, in the information-based definition, the bullwhip effect is a result of *one* decision maker, i.e., the unit in question. This decision maker observes demand and then makes order decisions based on various structural and economic factors. In the material-based definition, however, there are *three* decision effects involved. First, the sales data are determined by the actual demand and the on-hand inventory, where the latter is a result of the inventory decisions made in previous periods. Second, as in the information-based case, the unit makes order decisions based on structural and economic conditions. Third, the actual order receipts from the supplier are the result of the supplier’s previous production/stocking decisions, where the order receipts might not exactly equal the orders (e.g., production shortfall, transportation constraints, etc.). In view of these differences, we believe the information-based definition is more suitable for theoretical analysis purposes. However, we recognize the need for using the material-based definition as an empirical surrogate in some cases and thus include a discussion of the implications of such an approximation in §2.

Measuring the bullwhip effect in terms of aggregate data also prompts the question of potential aggregation biases in the measurement. In particular, Cachon et al. (2007) wrote, “Whether aggregation preserves or masks the bullwhip effect or production smoothing depends on the correlation of production and demand across the units being aggregate (firms, products, etc.) and on the particular causes of amplification in place” (p. 477). To give a rigorous treatment of this issue, we build on our individual-product inventory model to investigate the effect of data aggregation on the bullwhip measurement. We show that aggregating data over relatively long time periods can mask the bullwhip effect. In particular, when demand follows an autoregressive moving-average ARMA(1,1) model, we show that if the bullwhip ratio is greater than one to begin with, then the ratio decreases monotonically to one under time aggregation. When there is data aggregation across products, we show that the general insight obtained in our individual product analysis preserves, but the severity of the bullwhip can be masked in some illustrative cases. Our aggregation analysis is a generalization of Caplin (1985), where he considers aggregation across products with independently identically distributed (i.i.d.) demands and no capacity limits. We note that the insight obtained in our product aggregation analysis extends to location aggregation analysis as they are mathematically equivalent in our model.

Literature Review

The bullwhip effect phenomenon has been observed in many supply chains. For example, Hammond (1994) showed large fluctuations of weekly order quantities in Barilla’s pasta supply chain. Lee et al. (1997b) observed excess volatility in weekly orders in both Procter and Gamble’s diaper supply chain and Hewlett-Packard’s printer supply chain. Strong bullwhip effects were also

reported in various industries by Fransoo and Wouters (2000), Lai (2005), De Kok et al. (2005), and Waller et al. (2008). However, in a simulation study, Baganha and Cohen (1998) observed that order variability could be dampened due to the negatively correlated order stream generated by a downstream (s, S) inventory policy.

There also have been accounts of the bullwhip effect in the economics literature. These observations are primarily based on monthly or quarterly data aggregated across various products or firms. For example, high production volatility was found in the TV set industry (Holt et al. 1968), retail industry (Blinder 1981, Mosser 1991), automobile industry (Blanchard 1983, Kahn 1992), cement industry (Ghali 1987), and many other industries (Miron and Zeldes 1988, Fair 1989, Allen 1997). In these studies, researchers searched for explanations to reconcile the bullwhip effect with the classic production-smoothing theory, which posits that the motive for keeping inventory is to smooth production variability rather than to amplify it. One of the leading explanations is that production smoothing was missing because seasonality had been excluded from the data (e.g., Ghali 1987). Cachon et al. (2007) used monthly sales and inventory data from the U.S. Census Bureau to examine the bullwhip effect for various industries. They found that if seasonality is included in the measurement, production smoothing indeed exists in the retail industry and in some manufacturing industries, but not in the wholesale industry. In a recent study, using firm-level data, Bray and Mendelson (2012) found that two-thirds of the publicly traded companies bullwhip.

It is also useful to provide a brief survey of the main drivers for the bullwhip effect. An optimal order quantity from a rational decision maker is a response to both the supply/demand uncertainties and the cost structure of the situation. On the demand side, it is known in the literature that positively correlated demand coupled with long lead-time will amplify the bullwhip effect, while negatively correlated demand dampens it (Kahn 1987, Lee et al. 1997a, Chen and Lee 2009). Furthermore, certain commonly used demand forecast methods, such as moving average and exponential smoothing, will further increase order variability (Chen et al. 2000a, b). On the supply side, potential supply shortages will cause downstream stages to inflate orders and thus trigger the bullwhip (Lee et al. 1997a). The underlying cost structure also drives the order variability. For example, fixed ordering costs, such as full truckload and machine setup costs, will lead to large batch-orders and cause the bullwhip (Lee et al. 1997a, Cachon 1999, Chen et al. 2002). External cost shocks, such as promotional discounts, will induce forward-buying behavior, which again causes the bullwhip effect (Blinder 1986, Lee et al. 1997a). Conversely, explicit penalty costs for order variability will force the decision maker to smooth order quantities (Sobel 1969, Aviv 2007, Cantor and Katok 2012). Furthermore, a capacity constraint that truncates the order quantity is likely to smooth the order sequence; we will give a rigorous

treatment for this case in §2. In addition to these rational causes, there are also behavioral factors. For example, experimental studies have shown that the bounded rationality of human decision makers can cause excess order variability (Sterman 1989, Steckel et al. 2004, Croson and Donohue 2006). On the other hand, van Donselaar et al. (2010) found in an empirical study that store managers sometimes smoothed system-generated orders by moving orders from peak to nonpeak days.

The rest of this paper is organized as follows. Section 2 contains a general model analysis of the bullwhip measurement. In §3, we discuss the linkage between the bullwhip measure and supply chain cost performance as well as the data aggregation effect. Section 4 contains our concluding remarks. All proofs are presented in the appendix.

2. A General Model of Bullwhip Measurement

Let us consider a capacitated batch-order inventory system subject to a state-dependent stochastic demand with full backlog. Batch-ordering is quite common in practice because order quantities are often required to be rounded in the unit of pallet or carton case (a pallet usually contains multiple carton cases, and a carton case contains multiple sellable units). These batch-order units are usually predetermined either by the physical nature of the products, such as weight and/or cubic size, or by the economy of scale of the operations, so they cannot be easily altered.

Characterizing the optimal ordering policy for such a system is generally difficult. A plausible and commonly used heuristic in practice (such as Manhattan Associates and JDA Software) is to simply round the order quantity calculated from the capacitated system (assuming no batch-ordering) to the nearest multiple of the batch-order unit Q . In other words, the heuristic is a simple rounding of the modified base-stock policy, and we shall take this simple view throughout our analysis. We note that this heuristic can also be viewed as a close proxy to the modified (s, S) policy with $Q = S - s$. For a discussion of the modified (s, S) policy, we refer the readers to papers by Shaoxiang and Lambrecht (1996), Gallego and Scheller-Wolf (2000), and Shaoxiang (2004).

Without loss of generality, let us assume that the system capacity C is also a multiple of Q . The rounded order then takes values in the set of $\{0, Q, 2Q, \dots, C\}$. When there is no batch-ordering (i.e., $Q = 1$), it is known in the literature that a state-dependent modified base-stock policy is optimal for the capacitated system (Aviv and Federgruen 1997, Kapuściński and Tayur 1998). On the other hand, when there is no capacity limit (i.e., $C = \infty$), under the rounding heuristic, the system inventory position fluctuates around the state-dependent base-stock level S_t by $\pm Q/2$, which is mathematically equivalent to a system with a state-dependent (R_t, nQ) policy with $R_t = S_t - Q/2$.

Let us first examine the first step of the above two-step heuristic. In this step, a pre-rounding order quantity,

denoted by $\bar{O}_t$, is determined using the state-dependent modified base-stock policy. Let I_t be the inventory position after ordering in period t , S_t the base-stock level, and D_t the demand in period t . Then, according to the state-dependent modified base-stock policy, the order quantity $\bar{O}_t$ is given by

$$\begin{aligned}\bar{O}_t &= \min\{S_t - (I_{t-1} - D_{t-1}), C\} \\ &= \min\{S_t - S_{t-1} + D_{t-1} + S_{t-1} - I_{t-1}, C\} \\ &= \min\{O_t + \Delta_{t-1}, C\},\end{aligned}\quad (1)$$

where $O_t = S_t - S_{t-1} + D_{t-1}$ can be viewed as the would-be order quantity if there were no capacity constraint up to the current period, and $\Delta_{t-1} = S_{t-1} - I_{t-1}$ is the cumulative backorders (with $\Delta_{t-1} \geq 0$). In the above expression, we have implicitly assumed that both order quantities $\bar{O}_t$ and O_t can be negative. In other words, we have assumed excess inventory can be freely returned to the supplier. We will make this assumption throughout this paper for tractability. We note that allowing truncation at zero for order quantity would likely reduce the order variance, but such effect becomes negligible when the order mean is sufficiently greater than the order variance (as the chance of a negative order quantity becomes negligible). Justifications for this assumption can also be found in Lee et al. (2000), Aviv (2003, 2007), and Chen and Lee (2009).

From the above definitions, we can write the system bullwhip ratio as

$$\begin{aligned}\frac{\text{var}(\bar{O}_t)}{\text{var}(D_{t-1})} &= \frac{\text{var}(O_t)}{\text{var}(D_{t-1})} \cdot \frac{\text{var}(\bar{O}_t)}{\text{var}(O_t)} \\ &= \frac{\text{var}(S_t - S_{t-1} + D_{t-1})}{\text{var}(D_{t-1})} \cdot \frac{\text{var}(\bar{O}_t)}{\text{var}(O_t)}.\end{aligned}\quad (2)$$

The first term of the above expression captures the impact from the demand side through the covariance term between $S_t - S_{t-1}$ and D_{t-1} , while the second term captures the impact from the system capacity. Specifically, from expression (1), we see that $\bar{O}_t$ is the truncation of the sum of O_t and a random variable Δ_{t-1} . It is not immediately clear whether $\bar{O}_t$ is less variable than O_t or not. When $E\{O_t\} < C$, Δ_t visits the zero point infinitely many times. Let us examine a cycle that spans $\tau + 1$ periods between two consecutive zero-point visits, i.e., $\Delta_t = 0$, $O_{t+1} > C$, $O_{t+1} + O_{t+2} > 2C$, ..., $O_{t+1} + \dots + O_{t+\tau} > \tau C$, and $O_{t+1} + \dots + O_{t+\tau+1} \leq (\tau + 1)C$ (such that $\Delta_{t+\tau+1} = 0$). In this cycle, the truncated order quantity is given by

$$\bar{O}_{t+i} = \begin{cases} C & \text{if } 1 \leq i \leq \tau, \\ \sum_{i=1}^{\tau+1} O_{t+i} - \tau C & \text{if } i = \tau + 1. \end{cases}\quad (3)$$

It turns out that we can show $\sum_{i=1}^{\tau+1} \bar{O}_{t+i}^2 \leq \sum_{i=1}^{\tau+1} O_{t+i}^2$ (see the proof given in the appendix). Thus, it immediately follows from the ergodic theorem that $\text{var}(\bar{O}_t) \leq \text{var}(O_t)$.

Now let us examine the second step of the rounding heuristic. Let $\tilde{O}_t$ denote the rounded order quantity in period t . It is useful to determine the relationship between $\bar{O}_t$ and $\tilde{O}_t$. To facilitate the subsequent analysis, we assume that both D_t and $\bar{O}_t$ take only integer values, such that $\bar{O}_t$ can be expressed as $\bar{O}_t = m_t Q + q_t$, with m_t being a non-negative integer and $q_t \in \{0, 1, \dots, Q - 1\}$.

Let $\tilde{I}_t$ be the inventory position after order rounding. Define the rounding error as $\varepsilon_t = I_t - \tilde{I}_t$, where ε_t varies in the set of $\{-\lfloor Q/2 \rfloor + 1, \dots, 0, \dots, Q - \lfloor Q/2 \rfloor\}$ (where $\lfloor x \rfloor$ denotes the nearest integer below x).

The order quantity before rounding in period t is given by

$$\begin{aligned}\min\{S_t - (\tilde{I}_{t-1} - D_{t-1}), C\} \\ = \min\{S_t - I_{t-1} + D_{t-1} + I_{t-1} - \tilde{I}_{t-1}, C\} \\ = \min\{S_t - I_{t-1} + D_{t-1} + \varepsilon_{t-1}, C\}.\end{aligned}$$

Also, recall that

$$\bar{O}_t = \min\{S_t - I_{t-1} + D_{t-1}, C\} = m_t Q + q_t.$$

Thus, the rounded order quantity $\tilde{O}_t$ can be expressed as follows:

$$\tilde{O}_t = \begin{cases} C & \text{if } \bar{O}_t = C, \\ (m_t + 1)Q & \text{if } q_t + \varepsilon_{t-1} \geq Q/2, \text{ and } \bar{O}_t < C, \\ m_t Q & \text{if } q_t + \varepsilon_{t-1} < Q/2, \text{ and } \bar{O}_t < C. \end{cases}\quad (4)$$

To determine the variance of $\tilde{O}_t$, we need to establish the probability distribution of ε_{t-1} , which can be determined from the distribution of the Q -modular of the inventory position $\tilde{I}_{t-1}$.

Let $\hat{I}_{t-1} = (\tilde{I}_{t-1} \bmod Q)$, where $\hat{I}_{t-1}$ takes a value in $\{0, 1, \dots, Q - 1\}$. Because the rounded order quantity $\tilde{O}_t$ is always a multiple of Q , i.e., $(\tilde{O}_t \bmod Q) = 0$, then $\hat{I}_t$ is given by

$$\hat{I}_t = [(\hat{I}_{t-1} + \tilde{O}_t - D_{t-1}) \bmod Q] = [(\hat{I}_{t-1} - D_{t-1}) \bmod Q].$$

Now let $g_{t-1}(\cdot)$ denote the probability distribution of $(D_{t-1} \bmod Q)$ over the set $\{0, \dots, Q - 1\}$. Then the one-step transition matrix from $\hat{I}_{t-1}$ to $\hat{I}_t$ is given by

$$P_{t-1} = \begin{pmatrix} g_{t-1}(0) & g_{t-1}(Q-1) & \cdots & g_{t-1}(1) \\ g_{t-1}(1) & g_{t-1}(0) & \cdots & g_{t-1}(2) \\ \vdots & \vdots & \ddots & \vdots \\ g_{t-1}(Q-1) & g_{t-1}(Q-2) & \cdots & g_{t-1}(0) \end{pmatrix},$$

where $P_{t-1}(i, j) = \Pr\{\hat{I}_t = j \mid \hat{I}_{t-1} = i\}$ for $i, j \in \{0, 1, \dots, Q - 1\}$. We note that P_{t-1} given above is doubly stochastic because $\sum_{i=0}^{Q-1} g_{t-1}(i) = 1$.

PROPOSITION 1. Assume that O_t is a stationary process and $E\{O_t\} < C$. If P_t is irreducible for all t , then the stationary distribution for $\hat{I}_t$ is uniform over $\{0, 1, \dots, Q-1\}$. Furthermore, ε_t has a uniform distribution over $\{-\lfloor Q/2 \rfloor + 1, \dots, 0, \dots, Q - \lfloor Q/2 \rfloor\}$, and the system bullwhip ratio is given by

$$\frac{\text{var}(\bar{O}_t)}{\text{var}(D_{t-1})} = \frac{\text{var}(S_t - S_{t-1} + D_{t-1})}{\text{var}(D_{t-1})} \cdot \frac{\text{var}(\bar{O}_t)}{\text{var}(O_t)} \cdot \left(1 + \frac{E\{X_t(Q - X_t)\}}{\text{var}(\bar{O}_t)}\right), \quad (5)$$

where $X_t = (\bar{O}_t \bmod Q)$. Furthermore, the following holds: $(\text{var}(\bar{O}_t))/(\text{var}(O_t)) \leq 1$, with the equality holds at $C = \infty$; and $(E\{X_t(Q - X_t)\})/(\text{var}(\bar{O}_t)) \geq 0$, with the equality holds at $Q = 1$.

Equation (5) provides a general formula for the bullwhip ratio. From the formula, we can obtain an integrated view of the impacts of the demand process, system capacity, and batch-ordering on the bullwhip effect. When there is no batch-ordering (i.e., $Q = 1$), with an i.i.d. demand we have $S_t \equiv S_{t-1}$. Thus, in this case, it immediately follows that $\text{var}(\bar{O}_t)/\text{var}(D_{t-1}) = \text{var}(\bar{O}_t)/\text{var}(O_t) \leq 1$, so order variability is dampened. This confirms the intuition that imposing a finite capacity to the system would smooth the order variability. Chen and Lee (2009) showed that when there is no capacity limit (and no batch-ordering), under the martingale model of forecast evolution (MMFE) demand model, the bullwhip ratio is a function of the variance-covariance matrix of the evolving demand process. Furthermore, when there exists batch-ordering (i.e., $Q > 1$), the formula generalizes that of Caplin (1985) obtained in the i.i.d. demand setting with no capacity limit. In more general cases, the bullwhip ratio can be either greater or less than one, depending on the demand process characteristics (the first term), the tightness of the capacity constraint (the second term), and the magnitude of batch-ordering (the third term).

A few more discussion points are in order. First, the result of $\text{var}(\bar{O}_t) \leq \text{var}(O_t)$ can also be interpreted in the order-fulfillment setting where the downstream orders are truncated and backlogged according to the upstream fulfillment capacity. In this setting, the actual shipment corresponds to $\bar{O}_t$, the downstream order corresponds to O_t , and they share the same structure as given in (3). Thus, by a similar argument, we have $\text{var}(\text{shipment})/\text{var}(\text{order}) \leq 1$. Based on this observation, using the shipment (or order receipt) data as an empirical surrogate for the order data will understate the original order variance. This reinforces our early discussion that the material-based bullwhip effect is not the same as the information-based bullwhip effect. Empirically, Cachon et al. (2007, p. 465) calculated the variances of the orders and the inferred order receipts given in Hammond (1994) and found that the logged and first-differentiated order series has a variance of 1.28, whereas

the inferred order receipt data have a smaller variance of 1.25. Their observation is in accordance with our theory prediction.

Second, the amplifying effect of the third term in the formula can be viewed as an alternative proof of the order batching cause of the bullwhip effect discussed in Lee et al. (1997a). It also provides a theoretical explanation for a counterintuitive observation reported by Baganha and Cohen (1998). In a simulation study, they found that at the distribution center, order variability dampens because the orders generated by the downstream (s, S) policy are negatively correlated. They obtained this result assuming that the distribution center uses an order-up-to policy with no batch-ordering and no capacity limit, i.e., $Q = 1$ and $C = \infty$. Thus, from Proposition 1, the second and third terms of Equation (5) are both equal to one. They also approximated the demand process at the distribution center by an AR(1) process with a negative autocorrelation coefficient ρ in the simulation study; by setting $M = 1$, $\theta = 0$, and $\rho < 0$ in the formula given by Proposition 6 in §3, it is straightforward to verify that $\text{var}(O_t)/\text{var}(D_{t-1}) < 1$ under this demand process. Thus, the overall bullwhip ratio is less than one. If, however, the distribution center orders in large batches (as is often the case in real life), i.e., $Q \gg 1$, then according to Proposition 1, the overall bullwhip ratio (5) can be greater than one, and thus their simulation observation could be reversed. In fact, strong bullwhip effects are often observed at real-world distribution centers. For example, Lai (2005) found a large magnitude of bullwhip effect at the distribution center of a Spanish retailer. Waller et al. (2008) found similar effects in all 115 products at the distribution center of a major U.S. retailer. At the industry level, Cachon et al. (2007) observed that variability amplification exists in the wholesale industry with or without seasonality. Large batch-order size is a common practice at these supply chain locations because the operational efficiency there hinges upon economies of scale, such as full truckloads and volume discounts. Thus, the wholesalers, unlike the manufacturers who face finite manufacturing capacity, are less prone to production smoothing. Moreover, warehouses tend to be much less space constrained than retail stores. Hence, in the wholesale industry, the batch-ordering effect is more likely to dominate the finite capacity effect.

Finally, we make a few technical comments on Proposition 1. The uniformity result given in the proposition is essentially a generalization of those of Simon (1968) and Richards (1975) obtained under the i.i.d. demand and that of Browne and Zipkin (1991) obtained under the state-dependent stochastic demand with a static (R, Q) policy (as opposed to the state-dependent policy considered here). The root of this result lies in an observation by Feller (1971, p. 64) (see the proof in the appendix). The irreducibility of the transition matrix P_t is guaranteed if all states are communicated, which can be achieved by requiring $g_t(q) > 0$ for all t with $q \in \{1, \dots, Q-1\}$. A simple example would be to assume that there is a positive

probability for a single unit of demand in any given period, which is not at all restrictive.

2.1. Seasonal Demand Case I

Now let us consider a seasonal demand case with no capacity limit ($C = \infty$) and no batch-ordering ($Q = 1$). For simplicity, let us assume that an additive seasonality is present in the demand in each period. Let $s_0, \dots, s_{T-1}$ denote the normalized seasonality with a regular cycle of T periods, where s_0 corresponds to the seasonal factor of demand D_0 . By a “normalized” seasonality, we mean $\sum_{i=0}^{T-1} s_i = 0$. Let us also define the variability of seasonality as

$$V_s = \frac{1}{T} \sum_{i=0}^{T-1} s_i^2.$$

Let D_t (D'_t) and O_t (O'_t) denote the seasonal (deseasonalized) demand and order quantity in period t , respectively. Thus, the demand in a period can be expressed as

$$D_t = D'_t + s_{(t \bmod T)}.$$

Let us assume the replenishment leadtime is a constant of L periods. In this special case, the optimal base-stock level in a period is given by

$$S_t = S'_t + \sum_{i=0}^L s_{(t+i \bmod T)},$$

where S'_t is the state-dependent base-stock level for the deseasonalized demand process. Therefore, the seasonal order quantity in a period is given by

$$\begin{aligned} O_t &= S_t - (S_{t-1} - D_{t-1}) = S'_t - S'_{t-1} + D'_{t-1} + s_{(t+L \bmod T)} \\ &= O'_t + s_{(t+L \bmod T)}, \end{aligned}$$

where the last equality follows from relationship $O'_t = S'_t - S'_{t-1} + D'_{t-1}$.

PROPOSITION 2. Assume no capacity limit and no batch-ordering. The bullwhip ratio for a seasonal demand process is given by

$$\frac{\text{var}(O_t)}{\text{var}(D_{t-1})} = \frac{V_s + \text{var}(O'_t)}{V_s + \text{var}(D'_{t-1})} = 1 + \frac{\text{var}(O'_t) - \text{var}(D'_{t-1})}{V_s + \text{var}(D'_{t-1})},$$

where D'_t and O'_t are the deseasonalized demand and order quantity. Thus, if bullwhip exists, i.e., $\text{var}(O'_t) - \text{var}(D'_{t-1}) > 0$, the bullwhip ratio is monotone decreasing in V_s .

From the above result, we can see that if the variability of seasonality dominates the deseasonalized demand variability, i.e., $V_s \gg \text{var}(D'_{t-1})$, then including seasonality in the bullwhip measurement will have a stabilizing effect to make the ratio close to one. Empirically, Cachon et al. (2007) observed that when the material-based bullwhip ratio is greater than one, the material-based bullwhip measurement including seasonality is less than that excluding seasonality in most cases, which is in accordance with our theory prediction based on the information-based bullwhip ratio.

2.2. Seasonal Demand Case II

Consider another seasonal demand case where there is no batch-ordering but with a finite system capacity. Furthermore, we assume that the variability of seasonality is sufficiently larger than the deseasonalized demand variability (such that the latter is negligible). In other words, the demand process becomes deterministic, with a seasonal pattern. The demand in a period can thus be written as $D_t = \mu + s_{(t \bmod T)}$, with μ being the average demand level. Under this demand process, the optimal order quantity can be determined by solving the classic capacitated dynamic lot-sizing problem.

Assume that the system capacity $C \geq \mu$. Florian and Klein (1971) showed that the optimal order sequence to this problem satisfies either of the following two conditions: (1) $\bar{O}_t = D_{t+L}$, and (2) $\bar{O}_t$ belongs to a constrained sequence $\bar{O}_{t'}, \dots, \bar{O}_{t'+\tau}$ ($t' \leq t$ and $\tau > 0$) where all order levels are equal to C , except for one period with order level $\sum_{i=t'}^{t'+\tau} D_{t+i} - \tau C$. Note that our problem is a special case of Florian and Klein (1971) with no fixed order setup cost, so a positive order quantity should be placed whenever there is a demand requirement in L periods. Furthermore, note that the second condition has the same structure as given in (3). Thus, by a similar argument, we can show that for an arbitrary constrained sequence $\bar{O}_1, \dots, \bar{O}_{\tau+1}$, $\sum_{i=1}^{\tau+1} \bar{O}_i^2 \leq \sum_{i=1}^{\tau+1} D_{i+L}^2$. Then, the following result holds.

PROPOSITION 3. Assume a finite system capacity $C \geq \mu$ and no batch-ordering. Given a deterministic seasonal demand process, the bullwhip ratio under the optimal ordering sequence has the following property:

$$\frac{\text{var}(\bar{O}_t)}{\text{var}(D_{t-1})} \leq 1.$$

Furthermore, the ratio is increasing in C and reaches one at $C = \infty$.

This result provides a theoretical support for the production smoothing effect observed by Ghali (1987) and Cachon et al. (2007). In particular, Cachon et al. (2007) observed that when including seasonality in the material-based bullwhip measurement, variability dampens in the retail industry and some manufacturing industries. According to Proposition 3, the fact that the bullwhip ratios are below one in these industries is very likely due to two factors: (1) strong seasonality and (2) the prevalence of capacity constraints in the form of shelf-space/manufacturing capacity.

3. Cost Implications and Aggregation Analysis

In this section, we turn to the discussion of the linkage between the bullwhip measure and supply chain cost performance; we also discuss the data aggregation effect subsequently.

Consider a stylized supply chain model, where there is one retailer and one supplier. The retailer faces a random demand D_t and orders a quantity O_t from the supplier in each period t . In this simple supply chain, the retailer is the one who potentially creates the bullwhip. Clearly, the order variability from the retailer is a major cost driver for the supplier. As for the retailer, its cost might not depend directly on how much it bullwhips the upstream, other than maybe an eventual increase of wholesale prices from the supplier (which is difficult to quantify because it might depend on other factors). Below we discuss specifically the impact of the bullwhip effect on the supplier's order-fulfillment and inventory costs.

3.1. Order-Fulfillment Cost

Suppose the supplier fulfills the retailer's order every l ($l \geq 1$) periods and needs to determine the optimal order-fulfillment capacity K for this interval. The order-fulfillment capacity can be warehouse staffing or truck fleet size, for instance. Thus, the total orders to be fulfilled during the interval is given by $\sum_{t=1}^l O_t$. For ease of exposition, let us assume that both $\sum_{t=1}^l D_{t-1}$ and $\sum_{t=1}^l O_t$ are normally distributed, with variances being σ_D^2 and σ_O^2 , respectively. As a result, the retailer's bullwhip ratio measured at the interval of l periods is given by $\beta_l = \sigma_O^2 / \sigma_D^2$.

Let us assume that the supplier has sufficient inventory to meet all orders. The supplier's order-fulfillment capacity problem can then be modeled as a standard newsvendor problem:

$$\min_K \left\{ o \cdot E \left(K - \sum_{t=1}^l O_t \right)^+ + u \cdot E \left(\sum_{t=1}^l O_t - K \right)^+ \right\},$$

where o is the unit capacity overage cost, u is the unit capacity underage cost, and $x^+ = \max(x, 0)$. Because $\sum_{t=1}^l O_t$ follows a normal distribution, it is straightforward to show that the optimal order-fulfillment cost for the supplier is given by

$$C_o(\beta_l) = (o + u)\phi(z)\sigma_O = (o + u)\phi(z)\sigma_D\sqrt{\beta_l},$$

where $\phi(\cdot)$ is the standard normal density function and $z = \Phi^{-1}(u/(o + u))$, with $\Phi(\cdot)$ being the standard normal cumulative distribution function (see Porteus 2002). From the analysis, the following result immediately follows.

PROPOSITION 4. *Given the bullwhip ratio $\beta_l = \sigma_O^2 / \sigma_D^2$, the supplier's optimal order-fulfillment cost is given by $C_o(\beta_l) = \sqrt{\beta_l} \cdot C_o(1)$.*

The above proposition shows that the bullwhip measure has a direct linkage to the supplier's order-fulfillment cost. Specifically, the cost increases proportionally to the square-root of the bullwhip ratio. Suppose that the existing bullwhip ratio β_l is greater than one. If the supplier is able to manage its demand such that bullwhip equals one, for example, through means like vendor-managed inventory

(VMI), then the above proposition implies that the relative percentage of cost reduction for the supplier is given by $1 - C_o(1)/C_o(\beta_l) = (1 - 1/\sqrt{\beta_l}) \times 100\%$. In real life, however, when the supplier actively manages its demand through collaboration with the retailer, the existing bullwhip might not be completely eliminated (i.e., reduces to one). Thus, the above measure can be viewed as an upper bound of the relative performance improvement from programs such as VMI.

Aviv (2007, p. 789) has argued a similar point that reduced variability propagation can be beneficial to the supplier. However, it is worth noting that his cost analysis angle is different from ours. In our analysis, we focus on the supplier's order-fulfillment cost, which involves the capacity decisions such as warehouse staffing or truck fleet size, whereas Aviv (2007) emphasizes the production fluctuation cost resulting from the supplier's inventory production decisions. In some sense, our analysis captures the direct (order-fulfillment) cost to the supplier caused by the *incoming* bullwhip, while Aviv's cost analysis includes an additional (internal) bullwhip between the retailer's order and the supplier's production quantity.

Based on our analysis, we note that to correctly assess the supplier's cost performance, we need to measure the incoming bullwhip at the appropriate time unit, e.g., at the supplier's order-fulfillment interval l periods in this specific case. When the supplier fulfills the retailer's order as it receives (i.e., $l = 1$), the cost-relevant bullwhip ratio is the same as that measured at the retailer's order decision period (which we have analyzed in §2). If, however, the supplier adopts a longer order-fulfillment time interval ($l > 1$), then the cost-relevant bullwhip ratio is an aggregated measure over l periods. Therefore, it is important to understand the time aggregation effect of the bullwhip measure, which we will analyze below in §3.3.

In the above supply chain model, we have implicitly assumed that the upstream order-fulfillment interval is no less than the downstream order decision period. In many cases, however, the reverse is also true. For example, an individual consumer might buy diapers once every two weeks, but the store fulfills consumer demands on a daily basis. Or, a store might order from the retail distribution center weekly, but the retail distribution center fulfills all store orders daily. In these cases, because there are many consumers or stores, the aggregation of individual orders as a whole, which forms the incoming bullwhip to the supplying stage, might act like they come in frequently. Therefore, our insight remains valid—that is, the cost-relevant bullwhip measure is the incoming bullwhip measured at the upstream order-fulfillment interval.

Finally, besides the time unit for measurement, there is also an issue of using information-based bullwhip measure versus material-based bullwhip measure. As we have shown in §2, when the upstream has a shipment capacity limit, the manifested shipment variability will be less than the actual order variability. Thus, using the material-based

bullwhip measure might introduce additional biases in cost assessment.

3.2. Inventory Cost

The retailer's order variability can also have an impact on the supplier's inventory cost performance. Specifically, borrowing the inventory model from Chen and Lee (2009), let us assume that the supplier's inventory replenishment lead-time from an external source is L ($L \geq 0$) periods. Thus, the total demand during the inventory risk exposure period for the supplier is given by $\sum_{t=1}^{L+1} O_t$. For ease of exposition, let us assume that both $\sum_{t=1}^{L+1} D_{t-1}$ and $\sum_{t=1}^{L+1} O_t$ conditional on a common information set $\mathcal{F}_0$ (such as forecast updates) are normally distributed. Let $\sigma_{D|\mathcal{F}_0}^2 = \text{var}(\sum_{t=1}^{L+1} D_{t-1} | \mathcal{F}_0)$ and $\sigma_{O|\mathcal{F}_0}^2 = \text{var}(\sum_{t=1}^{L+1} O_t | \mathcal{F}_0)$. Thus, the bullwhip ratio measured at the interval of $L+1$ periods conditional on the information set $\mathcal{F}_0$ is given by $\beta_{L+1}(\mathcal{F}_0) = \sigma_{O|\mathcal{F}_0}^2 / \sigma_{D|\mathcal{F}_0}^2$. Note that this bullwhip measure captures the uncertainty propagation along the supply chain (as opposed to the unconditional variability propagation). When there is no information sharing between the retailer and the supplier, i.e., $\mathcal{F}_0$ is an empty set, the above uncertainty-based bullwhip ratio reduces to the traditional variability-based bullwhip measure.

According to the model of Chen and Lee (2009), the supplier's inventory optimization problem is given by

$$\min_{S_1} E \left\{ h \cdot \left(S_1 - \sum_{t=1}^{L+1} O_t \right)^+ + p \cdot \left(\sum_{t=1}^{L+1} O_t - S_1 \right)^+ \mid \mathcal{F}_0 \right\},$$

where h is the unit inventory holding cost, p is the unit stockout penalty cost, and S_1 is the base-stock level in period one. Because $\sum_{t=1}^{L+1} O_t$ conditional on $\mathcal{F}_0$ follows a normal distribution, similar to the order-fulfillment capacity problem, it is straightforward to show that the optimal inventory cost for the supplier is given by

$$\begin{aligned} C_I(\beta_{L+1}(\mathcal{F}_0)) &= (h + p)\phi(z)\sigma_{O|\mathcal{F}_0} \\ &= (h + p)\phi(z)\sigma_{D|\mathcal{F}_0}\sqrt{\beta_{L+1}(\mathcal{F}_0)}, \end{aligned}$$

where $z = \Phi^{-1}(p/(h + p))$. From the analysis, the following result immediately follows.

PROPOSITION 5. *Given the bullwhip ratio $\beta_{L+1}(\mathcal{F}_0) = \sigma_{O|\mathcal{F}_0}^2 / \sigma_{D|\mathcal{F}_0}^2$, the supplier's optimal inventory cost is given by $C_I(\beta_{L+1}(\mathcal{F}_0)) = \sqrt{\beta_{L+1}(\mathcal{F}_0)} \cdot C_I(1)$.*

The insight obtained in Proposition 5 is in accordance with that of Aviv (2007) and Chen and Lee (2009). Here we provide an explicit formula to link the supplier's inventory cost with the uncertainty-based bullwhip measure (conditional on the available information set $\mathcal{F}_0$). We comment that in this case, the traditional variability-based bullwhip measure remains relevant if the supplier has no access or does not utilize available demand information to improve its inventory replenishment decisions. We feel this case

could still apply to many small and medium size companies because they lack the resources in investing in advance information technology. Suppose that $\beta_{L+1}(\mathcal{F}_0)$ is greater than one. Similarly to what we discussed in the previous section, the above proposition implies that through programs like VMI, the relative percentage of inventory cost reduction for the supplier is directly dependent on the bullwhip ratio, with an upper bound given by $1 - C_I(1)/C_I(\beta_{L+1}(\mathcal{F}_0)) = (1 - 1/\sqrt{\beta_{L+1}(\mathcal{F}_0)}) \times 100\%$.

From the proposition, we again observe that the cost-relevant bullwhip effect for the supplier is the incoming bullwhip measured at the appropriate time unit: in this specific case, at the supplier's inventory risk exposure period of $L+1$ periods. In many supply chains, the decision periods are usually synchronized across the chain, e.g., members all order weekly. If the leadtimes become sufficiently short that we can treat it as if $L = 0$, then the bullwhip ratio measured at the retailer's order decision period (as studied in §2) is a proper measure to determine the impact on the supplier's inventory cost. Otherwise, we need to take into account the time aggregation effect over the proper time unit, which will be discussed in the following section.

3.3. Time Aggregation Analysis

Let us assume no capacity limit and no batch-ordering. Define the M -period aggregation of demand and order as $D_{t-1}^M = \sum_{\tau=0}^{M-1} D_{t-1+\tau}$ and $O_t^M = \sum_{\tau=0}^{M-1} O_{t+\tau}$, respectively. By the relationship $O_t = S_t - S_{t-1} + D_{t-1}$, it is easy to show that

$$O_t^M = S_{t+M-1} - S_{t-1} + D_{t-1}^M.$$

For most common demand models, it can be shown that $\lim_{M \rightarrow \infty} \text{var}(D_{t-1}^M) = \infty$. For example, it can be easily verified that this condition holds even for an AR(1) demand process with a negative autocorrelation coefficient. Thus, under this condition, the variance of the aggregated demand D_{t-1}^M will eventually dominate the finite variance of $S_{t+M-1} - S_{t-1}$ as M increases. Therefore, it is straightforward to show that $\lim_{M \rightarrow \infty} \text{var}(O_t^M) / \text{var}(D_{t-1}^M) = 1$. It is also straightforward to verify this result holds under finite capacity and batch-ordering.

To further illustrate the time aggregation effect, we obtained a data set from Rob Broekmeulen of Eindhoven University of Technology, which contains weekly sales and delivery data for six consumer products from a European retail store during a one-year period. (It would be ideal to use the same data set of Cachon et al. (2007) to illustrate, but their data is available only at the monthly level.) Table 2 presents the bullwhip ratios calculated at the time unit of weekly, biweekly, and every four weeks, where we used store delivery as a surrogate for store orders in the calculation. From the table, we can see that the bullwhip ratio tends to decrease as the aggregation period increases. When aggregation period increases from weekly to biweekly, bullwhip ratio decreases for most products except for apple-sauce and tea bags; these two anomalies are likely due

Table 2. Bullwhip ratio at a European retail store.

Product	Pack size	Weekly	Biweekly	Four-week
Applesauce	12	1.60	1.76	1.17
Mineral water	6	1.87	1.51	1.32
Peanut butter	12	1.50	1.25	1.08
Stroopwafels	15	1.31	1.23	1.17
Sugar	10	3.04	2.68	2.03
Tea bags	15	2.09	2.46	1.70
Aggregate		2.35	2.41	1.99

to the relatively small increment of aggregation period in this case. When aggregation period increases from weekly to every four weeks, the bullwhip ratio decreases for all products. In particular, for the peanut butter product, the bullwhip ratio at the four-week level is almost one.

PROPOSITION 6. Assume no capacity limit and no batch-ordering. Given a constant leadtime of L periods, under the ARMA(1, 1) demand model given by $D_t - \mu = \rho(D_{t-1} - \mu) + \epsilon_t - \theta\epsilon_{t-1}$, where $|\rho| < 1$, $|\theta| < 1$, and ϵ_t is i.i.d. normal with mean zero and standard deviation σ , the following holds:

$$\frac{\text{var}(O_t^M)}{\text{var}(D_{t-1}^M)} = 1 + \frac{2(\rho + \theta)(1 - \rho^{L+1})(1 - \rho^{L+2} + \theta\rho - \theta\rho^{L+1})}{(M/(1 - \rho^M))(1 - \rho^2)(1 + \theta)^2 - 2(\rho + \theta)(1 + \theta\rho)} \rightarrow 1, \text{ as } M \rightarrow \infty;$$

and the bullwhip ratio decreases monotonically in M when $\rho > 0$ and $\rho + \theta > 0$.

This result implies that insofar as an empirical demand process can be approximated by the AR(1), MA(1), or ARMA(1, 1) models (e.g., Erkip et al. 1990), the bullwhip ratio, if greater than one (i.e., when $\rho > 0$ and $\rho + \theta > 0$), will decrease monotonically to one as the aggregation period increases.

3.4. Product and Location Aggregation Analysis

Besides time aggregation, empirical data are also subject to product and location aggregation. Because location aggregation is mathematically equivalent to product aggregation, below we focus our analysis on product aggregation. Let us define the N -product aggregation of demand and order quantities as $D_{t-1}^N = \sum_{n=1}^N D_{t-1,n}$, $O_t^N = \sum_{n=1}^N O_{t,n}$, $\bar{O}_t^N = \sum_{n=1}^N \bar{O}_{t,n}$, and $\tilde{O}_t^N = \sum_{n=1}^N \tilde{O}_{t,n}$, where $D_{t-1,n}$ is the demand for product n and $O_{t,n}$, $\bar{O}_{t,n}$ and $\tilde{O}_{t,n}$ are the various order quantities for product n as defined in the individual product analysis. Also, define the aggregated base-stock level as $S_t^N = \sum_{n=1}^N S_{t,n}$. Let Q_n denote the batch-order unit for product n and C_n the capacity limit for product n . We can establish the following result under product aggregation.

PROPOSITION 7. If the inventory positions of the N products evolve according to an irreducible Markov chain, then the N -product aggregated bullwhip ratio is given by

$$\frac{\text{var}(\tilde{O}_t^N)}{\text{var}(D_{t-1}^N)} = \frac{\text{var}(S_t^N - S_{t-1}^N + D_{t-1}^N)}{\text{var}(D_{t-1}^N)} \cdot \frac{\text{var}(\bar{O}_t^N)}{\text{var}(O_t^N)} \cdot \left(1 + \frac{\sum_{n=1}^N E\{X_{t,n}(Q_n - X_{t,n})\}}{\text{var}(\bar{O}_t^N)}\right),$$

where $X_{t,n} = (\bar{O}_{t,n} \bmod Q_n)$.

This proposition is essentially a generalization of a product aggregation result of Caplin (1985) from products with i.i.d. demands and no capacity limits to the general case of products with state-dependent stochastic demands and finite capacity limits. Note that this formula does not require the demands across different products being independent. However, if demands are spatially independent across products, then by the result of Proposition 1 it is straightforward to show that $\text{var}(\bar{O}_t^N) \leq \text{var}(O_t^N)$, i.e., the smoothing effect of finite capacity preserves under product aggregation. Furthermore, if demands are positively spatially correlated across products (for example, aggregating across complementary product categories and/or across neighboring locations) such that $\lim_{N \rightarrow \infty} \text{var}(\bar{O}_t^N)/N = \infty$, then we can show that $\sum_{n=1}^N E\{X_{t,n}(Q_n - X_{t,n})\}/\text{var}(\bar{O}_t^N) \leq N \cdot (Q_{\max}^2/4)/\text{var}(\bar{O}_t^N) \rightarrow 0$, where $Q_{\max} = \max\{Q_1, \dots, Q_N\}$. Thus, in this case, the amplification effect due to batch-ordering will be masked under product aggregation.

Below, assuming away the capacity limit and batch-order constraint, we present two special demand processes that can also lead to the masking effect under product aggregation.

3.4.1. Special Case I: Common Seasonality. Let us first consider the case in which the products all share a common additive seasonality pattern, denoted by the normalized factors of $s_0, \dots, s_{T-1}$. As in §2.1, let $V_s = \sum_{i=0}^{T-1} s_i^2/T$ denote the variability of seasonality. Let $D'_{t,n}$ and $O'_{t,n}$ denote the deseasonalized demand and order quantity for product n , respectively. Thus, the demand for product n in a period can be expressed as

$$D_{t,n} = D'_{t,n} + \alpha_n \cdot s_{(t \bmod T)},$$

where the multiplicative factor α_n captures the heterogeneous magnitude of seasonality across the N products. Without loss of generality, let us assume $\alpha_n \geq 1$ for all n .

Assume the replenishment leadtime is a constant of L periods for all products. As shown in §2.1, the optimal order quantity for product n is given by

$$O_{t,n} = O'_{t,n} + \alpha_n \cdot s_{(t+L \bmod T)}. \quad (6)$$

PROPOSITION 8. Assume no capacity limit and no batch-ordering. Given N products with a common seasonal

pattern and spatially independent deseasonalized demands, the aggregate bullwhip ratio is given by

$$\frac{\text{var}(O_t^N)}{\text{var}(D_{t-1}^N)} = 1 + \frac{\sum_{n=1}^N \{\text{var}(O_{t,n}') - \text{var}(D_{t-1,n}')\}}{(\sum_{n=1}^N \alpha_n)^2 V_s + \sum_{n=1}^N \text{var}(D_{t,n}')} \rightarrow 1, \quad \text{as } N \rightarrow \infty.$$

From the above result, we can see that if the products under aggregation share a common seasonal profile (for example, the Christmas seasonality in the retail industry), then including seasonality in the measurement will drive the aggregated bullwhip ratio to one and thus mask the individual bullwhip.

3.4.2. Special Case II: AR(1) Demands. Consider another special case in which the demands of different products all belong to the AR(1) family (e.g., Erkip et al. 1990). Specifically, let us assume that the demand for product n is given by

$$D_{t,n} - \mu_n = \rho_n (D_{t-1,n} - \mu_n) + \epsilon_{t,n},$$

where $0 \leq \rho_n < 1$ and $\epsilon_{t,n}$ is i.i.d. normal with mean zero and standard deviation σ_n . Thus, we have $\text{var}(D_{t,n}) = \sigma_n^2 / (1 - \rho_n^2)$.

Assume the replenishment leadtime is a constant of L periods for all products. Then by setting $\rho = \rho_n$, $\theta = 0$ and $M = 1$ in the result of Proposition 6, we have the following:

$$\frac{\text{var}(O_{t,n})}{\text{var}(D_{t-1,n})} = 1 + \frac{2\rho_n(1 - \rho_n^{L+1})(1 - \rho_n^{L+2})}{1 - \rho_n} \geq 1. \quad (7)$$

PROPOSITION 9. Assume no capacity limit and no batch-ordering. Given N products with spatially independent demands that belong to the AR(1) family, if $\sum_{n=1}^N \rho_n \sigma_n^2 / \sum_{n=1}^N \sigma_n^2 / (1 - \rho_n) \rightarrow 0$ as $N \rightarrow \infty$, then the aggregate bullwhip ratio

$$\frac{\text{var}(O_t^N)}{\text{var}(D_{t-1}^N)} = 1 + \frac{\sum_{n=1}^N 2\rho_n \cdot (1 - \rho_n^{L+1}) / (1 - \rho_n) \cdot (1 - \rho_n^{L+2}) / (1 - \rho_n^2) \cdot \sigma_n^2}{\sum_{n=1}^N \sigma_n^2 / (1 - \rho_n^2)} \rightarrow 1 \quad \text{as } N \rightarrow \infty.$$

The condition of the above proposition can be satisfied, for example, when $\rho_n = 1/(n+1)$ and $\sigma_n = 1$ for all n , which means there is an increasing portion of the products with autocorrelation coefficient close to zero. In this case, the bullwhip ratios for individual products are all strictly greater than one, according to expression (7). Thus, this example shows that product aggregation can mask the severity of individual bullwhip. Another example is when $\rho_n = n/(n+1)$ and $\sigma_n = 1$ for all n . In this case, there is an increasing portion of products with autocorrelation coefficient approaching one. Because all products have bullwhip ratio strictly greater than one, this example again shows that product aggregation can mask the severity of individual bullwhip.

4. Conclusion

We have argued that it is important to delineate two commonly used bullwhip effect definitions in the literature. The original definition of the bullwhip effect by Lee et al. (1997a) is based on (1) information flow, (2) a single product, and (3) the order decision period. Many of the empirical studies, however, including that of Cachon et al. (2007), measure the bullwhip effect based on (1) material flow, (2) aggregated products, and (3) aggregated time to a month or longer. Hence, these empirical studies are essentially exploring the existence or the magnitude of “aggregated” bullwhip effects in material flow. Therefore, it might not have a direct bearing on the original bullwhip effect described by Lee et al. (1997a).

We have derived a general bullwhip formula that captures the effects of demand process characteristics, finite capacity, and batch-ordering. The multiplicative form of the formula makes it easy to separate each factor for econometrical analysis; we believe it can be used as a reference model for future empirical studies of the bullwhip effect. We have also demonstrated how this theoretical framework can be used to explain various empirical observations in the literature. Our analysis of the linkage between the bullwhip measure and the supply chain cost performance suggests that the bullwhip effect should be measured at the appropriate time unit for cost assessment purposes. For example, to assess its impact on the upstream warehouse staffing and truck fleet capacity costs, it should be measured at the upstream order-fulfillment time interval. To assess its impact on inventory-related costs, the bullwhip effect should be measured at the interval equal to the upstream inventory risk exposure period. The caveat here is that if there is forecast sharing between the downstream and the upstream stages, the bullwhip measure should be properly discounted to account for the actual demand uncertainty faced by the upstream stage (which is a conditional variance as opposed to the total variability captured by the traditional bullwhip measure).

Building on this framework, we have further shown how data aggregation can affect the bullwhip measurement. Although not surprising, the masking effect of time aggregation has an interesting interpretation. Recall that we have shown that the upper bound of the relative percentage of cost reduction from programs like VMI is directly linked to the bullwhip ratio (see §§3.1 and 3.2). When the upstream stage reduces its order-fulfillment interval, reorder period or replenishment leadtime, the bullwhip ratio, measured at the shorter time interval, would increase; as a result, the relative performance improvement of a VMI program increases, even though the absolute performance improvement of such a program is likely to be smaller compared to the case without such time reduction. Our aggregation analysis also suggests that the bullwhip effect tends to be less dramatic at the firm or industry level than at the individual product level. Given the fact that most financial planning and capital investment decisions are based on firm-level aggregate

data on a quarterly or even yearly basis, the severity of the product-level bullwhip effect is very likely to be overlooked by the aggregate planners.

Appendix

PROOF (PROPOSITION 1). The proof is divided into two parts. In the first part, we prove $\text{var}(\tilde{O}_t) \leq \text{var}(O_t)$. For any cycle such that $\Delta_t = 0$, $O_{t+1} > C$, $O_{t+1} + O_{t+2} > 2C$, ..., $O_{t+1} + \dots + O_{t+\tau} > \tau C$, and $O_{t+1} + \dots + O_{t+\tau+1} \leq (\tau + 1)C$ (such that $\Delta_{t+\tau+1} = 0$), we have

$$\begin{aligned} & \sum_{i=1}^{\tau+1} \tilde{O}_{t+i}^2 - \sum_{i=1}^{\tau+1} O_{t+i}^2 \\ &= \tau C^2 + \left[\sum_{i=1}^{\tau+1} O_{t+i} - \tau C \right]^2 - \sum_{i=1}^{\tau+1} O_{t+i}^2 \\ &= (\tau + 1)C^2 + \sum_{i=1}^{\tau+1} (O_{t+i} - C)^2 + 2C \left[\sum_{i=1}^{\tau+1} O_{t+i} - (\tau + 1)C \right] \\ &\quad + 2 \sum_{i=1}^{\tau+1} \sum_{j=1}^{i-1} (O_{t+i} - C)(O_{t+j} - C) - \sum_{i=1}^{\tau+1} O_{t+i}^2 \\ &= 2 \sum_{i=1}^{\tau+1} \sum_{j=1}^{i-1} (O_{t+i} - C)(O_{t+j} - C). \end{aligned}$$

Let us show that $\sum_{i=1}^{\tau+1} \sum_{j=1}^{i-1} (O_{t+i} - C)(O_{t+j} - C) < 0$. Note that we have $\sum_{i=1}^k O_{t+i} > kC$ for $k = 1, \dots, \tau$, and $\sum_{i=1}^{\tau+1} O_{t+i} \leq (\tau + 1)C$. Now let us prove by induction. For $\tau = 1$, we have $(O_{t+1} - C)(O_{t+2} - C) < 0$ because $O_{t+1} > C$ and $O_{t+1} + O_{t+2} \leq 2C$ (which implies $O_{t+2} < C$). Now assume the result holds true for any cycle length up to τ . For the case of $\tau + 1$, we have $O_{t+\tau+1} < C$. Because $O_{t+1} > C$, we can always find $k = \max\{i \mid O_{t+i} > C, 1 \leq i \leq \tau\}$. Now by the definition, $O_{t+k} > C$ implies $\sum_{i \neq k} O_{t+i} < \tau C$. Therefore, we can always find $l = \min\{j \mid \sum_{i \neq k, 1 \leq i \leq j} O_{t+i} \leq (j - 1)C, k + 1 \leq j \leq \tau + 1\}$. Given k and l , we have $O_{t+i} \leq C$ for $l + 1 \leq i \leq \tau + 1$, and $\sum_{i \neq k, 1 \leq i \leq j} O_{t+i} > (j - 1)C$ for $1 \leq j \leq l - 1$. Hence,

$$\begin{aligned} & \sum_{i=1}^{\tau+1} \sum_{j=1}^{i-1} (O_{t+i} - C)(O_{t+j} - C) \\ &= \sum_{i=1}^l \sum_{j=1}^{i-1} (O_{t+i} - C)(O_{t+j} - C) \\ &\quad + \sum_{i=l+1}^{\tau+1} \sum_{j=1}^{i-1} (O_{t+i} - C)(O_{t+j} - C) \\ &= \sum_{i \neq k, i=1}^l \sum_{j \neq k, j=1}^{i-1} (O_{t+i} - C)(O_{t+j} - C) \\ &\quad + (O_{t+k} - C) \left(\sum_{j \neq k, j=1}^l O_{t+j} - (l - 1)C \right) \\ &\quad + \sum_{i=l+1}^{\tau+1} (O_{t+i} - C) \left(\sum_{j=1}^{i-1} O_{t+j} - (i - 1)C \right) \leq 0, \end{aligned}$$

where the first term is negative because the order cycle of $(i \neq k, 1 \leq i \leq l)$ satisfies the induction assumption of case $l - 1$ ($\leq \tau$), the second term is negative because by definition $O_{t+k} > C$ and

$\sum_{j \neq k, 1 \leq j \leq l} O_{t+j} \leq (l - 1)C$, and the third term is negative because by definition $O_{t+i} \leq C$ for $i \geq l + 1 > k$ and $\sum_{j=1}^{i-1} O_{t+j} > (i - 1)C$ for any $i < \tau$. This completes the induction proof.

Thus, we conclude that $\sum_{i=1}^{\tau+1} \tilde{O}_{t+i}^2 < \sum_{i=1}^{\tau+1} O_{t+i}^2$. Also, it is straightforward to verify that $\sum_{i=1}^{\tau+1} \tilde{O}_{t+i} = \sum_{i=1}^{\tau+1} O_{t+i}$. Because O_t is a stationary process, by the ergodic theorem, we conclude that $E\{\tilde{O}_t^2\} \leq E\{O_t^2\}$ and $E\{\tilde{O}_t\} = E\{O_t\}$. Therefore, $\text{var}(\tilde{O}_t) - \text{var}(O_t) = E\{\tilde{O}_t^2\} - E\{O_t^2\} \leq 0$. This concludes the proof of the first part.

In the second part, we prove the result that $\text{var}(\tilde{O}_t)/\text{var}(O_t) = 1 + E\{X_t(Q - X_t)\}/\text{var}(\tilde{O}_t) \geq 1$. Because P_t is finite and irreducible for all t , there exists a unique stationary distribution for $\hat{I}_t$, which is a uniform distribution because P_t is doubly stochastic. We know that I_t could be either S_t or $I_{t-1} + C - D_{t-1}$. When $I_t = S_t$, we have

$$\begin{aligned} \varepsilon_t \bmod Q &= (S_t - \tilde{I}_t) \bmod Q \\ &= ((S_t \bmod Q) - \tilde{I}_t) \bmod Q. \end{aligned}$$

Because $\hat{I}_t$ has a uniform distribution and is independent of S_t , after the modular operation we obtain that $(\varepsilon_t \bmod Q)$ also has a uniform distribution (see Feller 1971, p. 64). Note that ε_t has the same distribution as $(\varepsilon_t \bmod Q)$. Hence, ε_t is uniformly distributed when $I_t = S_t$. When $I_t = I_{t-1} + C - D_{t-1}$, we have

$$\begin{aligned} \varepsilon_t \bmod Q &= (I_t - \tilde{I}_t) \bmod Q \\ &= (I_{t-1} + C - D_{t-1} - \tilde{I}_t) \bmod Q \\ &= (I_{t-1} + C - \tilde{I}_{t-1} - \tilde{O}_{t-1}) \bmod Q \\ &= \varepsilon_{t-1} \bmod Q. \end{aligned}$$

Hence, we conclude that the stationary distribution of ε_t is uniformly distributed.

Now note that $E(\tilde{O}_t) = C \cdot \Pr\{\tilde{O}_t = C\} + \sum_{\tilde{O}_t < C} (m_t Q + q_t) \Pr\{\tilde{O}_t = m_t Q + q_t\}$. Similarly, we have

$$\begin{aligned} E(\tilde{O}_t) &= C \cdot \Pr\{\tilde{O}_t = C\} + \sum_{\tilde{O}_t < C} \left[(m_t + 1)Q \cdot \frac{q_t}{Q} + m_t Q \cdot \frac{Q - q_t}{Q} \right] \\ &\quad \cdot \Pr\{\tilde{O}_t = m_t Q + q_t\} \\ &= C \cdot \Pr\{\tilde{O}_t = C\} + \sum_{\tilde{O}_t < C} (m_t Q + q_t) \Pr\{\tilde{O}_t = m_t Q + q_t\}, \end{aligned}$$

where the first equality follows from Equation (4) and the uniform distribution of ε_{t-1} . Hence, we have $E(\tilde{O}_t) = E(O_t)$. Therefore,

$$\begin{aligned} \text{var}(\tilde{O}_t) - \text{var}(O_t) &= E(\tilde{O}_t^2) - E(O_t^2) \\ &= \sum_{\tilde{O}_t < C} \left[(m_t + 1)^2 Q^2 \cdot \frac{q_t}{Q} + m_t^2 Q^2 \cdot \frac{Q - q_t}{Q} \right] \Pr\{\tilde{O}_t = m_t Q + q_t\} \\ &\quad - \sum_{\tilde{O}_t < C} (m_t Q + q_t)^2 \Pr\{\tilde{O}_t = m_t Q + q_t\} \\ &= \sum_{\tilde{O}_t < C} q_t (Q - q_t) \Pr\{\tilde{O}_t = m_t Q + q_t\} \\ &= \sum_{x=0}^{Q-1} x(Q - x) \Pr\{x = (\tilde{O}_t \bmod Q), \tilde{O}_t < C\} \\ &= E\{X_t(Q - X_t)\}, \end{aligned}$$

where the second equality follows from Equation (4) and the uniform distribution of ε_{t-1} , and the last equality follows from the

definition of $X_t = (\bar{O}_t \bmod Q)$ and the fact that $(C \bmod Q) = 0$. Hence,

$$\frac{\text{var}(\bar{O}_t)}{\text{var}(\bar{O}_t)} = 1 + \frac{E\{X_t(Q - X_t)\}}{\text{var}(\bar{O}_t)} \geq 1.$$

From the above expression, it is easy to see that when $Q = 1$, $\text{var}(\bar{O}_t) = \text{var}(\bar{O}_t)$. Combining the above result with Equation (2), we obtain the system bullwhip ratio. $\square$

PROOF (PROPOSITION 2). It is straightforward to verify that the order variability with seasonality included is given by

$$\begin{aligned} \text{var}(O_t) &= \frac{1}{T} \sum_{j=0}^{T-1} E\{(s_{(t+j+L \bmod T)} + O'_{t+j} - E\{O'_{t+j}\})^2\} \\ &= \frac{1}{T} \sum_{j=0}^{T-1} \{s_{(t+j+L \bmod T)}^2 + \text{var}(O'_{t+j})\} \\ &= V_s + \text{var}(O'_t), \end{aligned}$$

where the last equality follows from the fact that O'_t is stationary. Similarly, the demand variability with seasonality included can be shown as $\text{var}(D_{t-1}) = V_s + \text{var}(D'_{t-1})$. Hence, we arrive at the desired result. $\square$

PROOF (PROPOSITION 3). By a similar argument given in the proof of Proposition 1, for any constrained sequence $\bar{O}_1, \dots, \bar{O}_{\tau+1}$, we have $\sum_{i=1}^{\tau+1} \bar{O}_i^2 \leq \sum_{i=1}^{\tau+1} D_{i+L}^2$. According to the two conditions of the optimal order sequence during a seasonal cycle T , we conclude that $\sum_{i=1}^T \bar{O}_{t+i}^2 \leq \sum_{i=1}^T D_{t+L+i}^2$. Also, because $C \geq \mu$, we have $\sum_{i=1}^T \bar{O}_{t+i} = \sum_{i=1}^T D_{t+L+i} = T\mu$. Hence, we have

$$\begin{aligned} \text{var}(\bar{O}_t) &= \frac{1}{T} \sum_{i=1}^T (\bar{O}_{t+i} - \mu)^2 = \frac{1}{T} \sum_{i=1}^T \bar{O}_{t+i}^2 - \mu^2 \\ &\leq \frac{1}{T} \sum_{i=1}^T D_{t+L+i}^2 - \mu^2 = \text{var}(D_{t-1}). \end{aligned}$$

For the monotonicity result, let $\bar{O}'_t$ denote the order quantity subject to a capacity limit C' , with $C' < C$. By the optimal order sequence property of $\bar{O}'_t$ and $\bar{O}_t$, we can conclude that either (1) $\bar{O}'_t = \bar{O}_t = D_{t+L}$ or (2) $\bar{O}'_t$ belongs to a constrained sequence where all order levels are equal to C' except for one period with order level $\sum_{i=t'}^{t'+\tau} D_{i+L} - \tau C' = \sum_{i=t'}^{t'+\tau} \bar{O}_i - \tau C'$. Thus, we can effectively treat the optimal order sequence $\bar{O}_t$ as the demand requirement. By the same argument shown above, we have $\text{var}(\bar{O}'_t) \leq \text{var}(\bar{O}_t)$. Hence, the bullwhip ratio $\text{var}(\bar{O}_t)/\text{var}(D_{t-1})$ is increasing in C , and it reaches one at $C = \infty$. $\square$

PROOF (PROPOSITION 4). The result follows immediately from the proceeding analysis with some algebra. $\square$

PROOF (PROPOSITION 5). The result follows immediately from the proceeding analysis with some algebra. $\square$

PROOF (PROPOSITION 6). From the ARMA(1, 1) demand model, by some algebra, it is easy to verify that

$$\begin{aligned} \text{var}(D_{t-1}^M) &= \frac{M(1-\rho^2)(1+\theta)^2 - 2(\rho+\theta)(1+\theta\rho)(1-\rho^M)}{(1-\rho^2)(1-\rho)^2} \cdot \sigma^2. \end{aligned}$$

By Proposition 1 of Chen and Lee (2009), it is also easy to verify that

$$\begin{aligned} \text{var}(O_t^M) - \text{var}(D_{t-1}^M) &= \frac{2(\rho+\theta)(1-\rho^{L+1})(1-\rho^{L+2}+\theta\rho-\theta\rho^{L+1})(1-\rho^M)}{(1-\rho^2)(1-\rho)^2} \cdot \sigma^2. \end{aligned}$$

Hence, we have

$$\begin{aligned} \frac{\text{var}(O_t^M)}{\text{var}(D_{t-1}^M)} &= 1 + \frac{2(\rho+\theta)(1-\rho^{L+1})(1-\rho^{L+2}+\theta\rho-\theta\rho^{L+1})}{M/(1-\rho^M)(1-\rho^2)(1+\theta)^2 - 2(\rho+\theta)(1+\theta\rho)}. \end{aligned}$$

Note that $M/(1-\rho^M)$ is increasing in M for $\rho \geq 0$. Also, $(\rho+\theta)(1-\rho^{L+2}+\theta\rho-\theta\rho^{L+1}) \geq 0$ if and only if $\rho+\theta \geq 0$. Hence, it follows that the order variance amplification ratio is monotonically decreasing in M when $\rho \geq 0$ and $\rho+\theta \geq 0$. $\square$

PROOF (PROPOSITION 7). Let $\hat{\mathbf{I}}_t = [\hat{I}_{t,1}, \dots, \hat{I}_{t,N}]^T$ denote the Q -modular inventory positions for the N products, where $\hat{I}_{t,n} = (\bar{I}_{t,n} \bmod Q_n)$ for $n = 1, \dots, N$. Now let $g_t(x_1, \dots, x_N)$ denote the joint probability distribution of $x_1 = (D_{t,1} \bmod Q_1), \dots, x_N = (D_{t,N} \bmod Q_N)$ over the Cartesian set $\mathcal{A} = \{0, \dots, Q_1-1\} \times \dots \times \{0, \dots, Q_N-1\}$. Also let P_{t-1}^N be the one-step transition matrix from $\hat{\mathbf{I}}_{t-1}$ to $\hat{\mathbf{I}}_t$. We can show that

$$\Pr\{\hat{\mathbf{I}}_t | \hat{\mathbf{I}}_{t-1}\} = g_{t-1}(z_1, \dots, z_N),$$

with $z_n = [(\hat{I}_{t-1,n} - \hat{I}_{t,n}) \bmod Q_n]$ for $n = 1, \dots, N$. Note that for any pairs of $\hat{\mathbf{I}}_{t-1}$ and $\hat{\mathbf{I}}_t$, there is a unique $g_{t-1}(z_1, \dots, z_N)$. Therefore, for any given vector $\hat{\mathbf{I}}_t \in \mathcal{A}$

$$\Pr\{\hat{\mathbf{I}}_t | \text{for all } \hat{\mathbf{I}}_{t-1} \in \mathcal{A}\} = \sum_{(z_1, \dots, z_N) \in \mathcal{A}} g_{t-1}(z_1, \dots, z_N) = 1.$$

Thus, the transition matrix P_{t-1}^N is doubly stochastic. Hence, $\hat{\mathbf{I}}_t$ is a uniform distribution over set $\mathcal{A}$. By the analogous argument used for Proposition 1, it is easy to show that $\varepsilon_{t,n}$ is also uniformly distributed.

We want to show that $\text{var}(\bar{O}_t^N) - \text{var}(\bar{O}_t^N) = \sum_{n=1}^N \{\text{var}(\bar{O}_{t,n}) - \text{var}(\bar{O}_{t,n})\}$. Thus, it suffices to show that for any pair of product locations i, j ($i \neq j$), $\text{cov}(\bar{O}_{t,i}, \bar{O}_{t,j}) = \text{cov}(\bar{O}_{t,i}, \bar{O}_{t,j})$. From Proposition 1, we know that $E\{\bar{O}_{t,n}\} = E\{\bar{O}_{t,n}\}$ for any n . So it suffices to show that $E\{\bar{O}_{t,i} \cdot \bar{O}_{t,j}\} = E\{\bar{O}_{t,i} \cdot \bar{O}_{t,j}\}$. Let $\bar{O}_{t,i} = m_{t,i}Q_i + q_{t,i}$ and $\bar{O}_{t,j} = m_{t,j}Q_j + q_{t,j}$. By utilizing the property of uniform distribution of $\varepsilon_{t-1,i}$ and $\varepsilon_{t-1,j}$ and the Equation (4), we have

$$\begin{aligned} E\{\bar{O}_{t,i} \cdot \bar{O}_{t,j}\} &= C_i C_j \Pr\{\bar{O}_{t,i} = C_i, \bar{O}_{t,j} = C_j\} \\ &\quad + C_i E\{\bar{O}_{t,j} | \bar{O}_{t,i} = C_i, \bar{O}_{t,j} < C_j\} \Pr\{\bar{O}_{t,i} = C_i, \bar{O}_{t,j} < C_j\} \\ &\quad + C_j E\{\bar{O}_{t,i} | \bar{O}_{t,j} = C_j, \bar{O}_{t,i} < C_i\} \Pr\{\bar{O}_{t,j} = C_j, \bar{O}_{t,i} < C_i\} \\ &\quad + \sum_{\bar{O}_{t,i} < C_i} \sum_{\bar{O}_{t,j} < C_j} \left[(m_{t,i}+1)Q_i \cdot \frac{q_{t,i}}{Q_i} \cdot (m_{t,j}+1)Q_j \cdot \frac{q_{t,j}}{Q_j} \right. \\ &\quad \quad \quad + (m_{t,i}+1)Q_i \cdot \frac{q_{t,i}}{Q_i} \cdot m_{t,j}Q_j \cdot \frac{Q_j - q_{t,j}}{Q_j} \\ &\quad \quad \quad + m_{t,i}Q_i \cdot \frac{Q_i - q_{t,i}}{Q_i} \cdot (m_{t,j}+1)Q_j \cdot \frac{q_{t,j}}{Q_j} \\ &\quad \quad \quad \left. + m_{t,i}Q_i \cdot \frac{Q_i - q_{t,i}}{Q_i} \cdot m_{t,j}Q_j \cdot \frac{Q_j - q_{t,j}}{Q_j} \right] \\ &\quad \cdot \Pr\{\bar{O}_{t,i} = m_{t,i}Q_i + q_{t,i}, \bar{O}_{t,j} = m_{t,j}Q_j + q_{t,j}\} \end{aligned}$$

$$\begin{aligned}
&= C_i C_j \Pr\{\bar{O}_{t,i} = C_i, \bar{O}_{t,j} = C_j\} \\
&\quad + C_i E\{\bar{O}_{t,j} \mid \bar{O}_{t,i} = C_i, \bar{O}_{t,j} < C_j\} \Pr\{\bar{O}_{t,i} = C_i, \bar{O}_{t,j} < C_j\} \\
&\quad + C_j E\{\bar{O}_{t,i} \mid \bar{O}_{t,j} = C_j, \bar{O}_{t,i} < C_i\} \Pr\{\bar{O}_{t,j} = C_j, \bar{O}_{t,i} < C_i\} \\
&\quad + \sum_{\bar{O}_{t,i} < C_i} \sum_{\bar{O}_{t,j} < C_j} \left[(m_{t,i} Q_i + q_{t,i}) \cdot (m_{t,j} Q_j + q_{t,j}) \right] \\
&\quad \cdot \Pr\{\bar{O}_{t,i} = m_{t,i} Q_i + q_{t,i}, \bar{O}_{t,j} = m_{t,j} Q_j + q_{t,j}\} \\
&= E\{\bar{O}_{t,i} \cdot \bar{O}_{t,j}\}.
\end{aligned}$$

Hence,

$$\begin{aligned}
\frac{\text{var}(\bar{O}_t^N)}{\text{var}(\bar{O}_t^N)} &= 1 + \frac{\sum_{n=1}^N \{\text{var}(\bar{O}_{t,n}) - \text{var}(\bar{O}_{t,n})\}}{\text{var}(\bar{O}_t^N)} \\
&= 1 + \frac{\sum_{n=1}^N E\{X_{t,n}(Q - X_{t,n})\}}{\text{var}(\bar{O}_t^N)} \geq 1,
\end{aligned}$$

where $X_{t,n} = (\bar{O}_{t,n} \bmod Q_n)$ and the last equality follows from the individual product analysis of Proposition 1. $\square$

PROOF (PROPOSITION 8). By Equation (6), it is straightforward to verify that

$$\begin{aligned}
\text{var}(O_t^N) &= \frac{1}{T} \sum_{j=0}^{T-1} E \left\{ \left(\sum_{n=1}^N \alpha_n \cdot s_{(t+j+L \bmod T)} \right. \right. \\
&\quad \left. \left. + \sum_{n=1}^N [O'_{t+j,n} - E\{O'_{t+j,n}\}] \right)^2 \right\} \\
&= \frac{1}{T} \sum_{j=0}^{T-1} \left\{ \left(\sum_{n=1}^N \alpha_n \right)^2 \cdot s_{(t+j+L \bmod T)}^2 + \sum_{n=1}^N \text{var}(O'_{t+j,n}) \right\} \\
&= \left(\sum_{n=1}^N \alpha_n \right)^2 \cdot V_s + \sum_{n=1}^N \text{var}(O'_{t,n}),
\end{aligned}$$

where the last equality follows from the fact that $O'_{t,n}$ is stationary. Similarly, the variability of demand can be shown as $\text{var}(D_{t-1}^N) = (\sum_{n=1}^N \alpha_n)^2 V_s + \sum_{n=1}^N \text{var}(D'_{t,n})$. Note that $\sum_{n=1}^N \alpha_n \geq N$. Hence, we have

$$\begin{aligned}
\frac{\text{var}(O_t^N)}{\text{var}(D_{t-1}^N)} &= 1 + \frac{\sum_{n=1}^N \{\text{var}(O'_{t,n}) - \text{var}(D'_{t-1,n})\}}{\left(\sum_{n=1}^N \alpha_n \right)^2 V_s + \sum_{n=1}^N \text{var}(D'_{t,n})} \\
&\leq 1 + \frac{N \cdot \max_{1 \leq n \leq N} |\text{var}(O'_{t,n}) - \text{var}(D'_{t-1,n})|}{N^2 \cdot V_s + \sum_{n=1}^N \text{var}(D'_{t,n})} \rightarrow 1,
\end{aligned}$$

as $N \rightarrow \infty$. $\square$

PROOF (PROPOSITION 9). It is easy to show that

$$\begin{aligned}
\frac{\text{var}(O_t^N)}{\text{var}(D_{t-1}^N)} &= 1 + \frac{\sum_{n=1}^N 2\rho_n \cdot (1 - \rho_n^{L+1}) / (1 - \rho_n) \cdot (1 - \rho_n^{L+2}) / (1 - \rho_n^2) \cdot \sigma_n^2}{\sum_{n=1}^N \sigma_n^2 / (1 - \rho_n^2)} \\
&\leq 1 + \frac{4(L+1)(L+2) \sum_{n=1}^N \rho_n \sigma_n^2}{\sum_{n=1}^N \sigma_n^2 / (1 - \rho_n)},
\end{aligned}$$

where the inequality follows from the fact that $(1 - \rho_n^{L+1}) / (1 - \rho_n) = 1 + \rho_n + \dots + \rho_n^L \leq L + 1$ and $(1 - \rho_n^{L+2}) / (1 - \rho_n^2) = (1 + \rho_n + \dots + \rho_n^{L+1}) / (1 + \rho_n) \leq L + 2$. Therefore, the aggregated bullwhip ratio will approach one if $\sum_{n=1}^N \rho_n \sigma_n^2 / \sum_{n=1}^N \sigma_n^2 / (1 - \rho_n) \rightarrow 0$ as $N \rightarrow \infty$. $\square$

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Li Chen is an assistant professor at the Fuqua School of Business at Duke University. His research interests include information and optimal control in operations and supply chain management, behavioral decision making, and sustainable business operations.

Hau L. Lee is the Thoma Professor of Operations, Information and Technology at the Graduate School of Business of Stanford University. His research interests include global supply chain management, sustainability, and responsible business.

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